

Computer Architecture Notes: MS-DOS & The System Path

1. What is MS-DOS (Command Prompt)?

- **The Blind Text Messenger:** MS-DOS (and the modern Command Prompt `cmd`) has no eyes. It cannot see your desktop, your shortcuts, or your visual folders.
- **Text Commands Only:** When you type a word like `java` or `python`, MS-DOS does not know what the programming language is. It assumes you are asking it to run a physical executable file named `java.exe` or `python.exe` hidden on your hard drive.
- **The Lazy Shortcut:** Humans can type just `java` instead of `java.exe`. MS-DOS automatically appends the `.exe` extension in its own background process. It also **does not care about capital letters** (`java`, `Java`, and `JAVA` all mean the exact same thing to Windows).

2. The Current Folder Rule

Before looking anywhere else on the computer, MS-DOS always looks for your specified file in the **exact folder you are currently standing in** inside the command prompt windows (e.g., `C:\Users\YourName`).

- If the `.exe` file is present there, it runs it immediately.
- If it is not present in that folder, it moves on to trigger the **Path search**.

3. What is the "Path"?

- **The Master Cheat Sheet:** The Path is **one single global environment variable** shared by the entire Windows operating system. All programming languages (Python, JavaScript, Java) must share this same single list.
- **An Unlabelled List of Folder URLs:** The Path is **not** an organized object with clear labels like *Name: Java, URL:* Instead, it is just a flat, blind text string of folder addresses separated strictly by semicolons (`;`).
- **Why Multiple Folders Exist:** Different companies build different software. Python, Node.js, and Java all insist on living in their own isolated folders (sandboxes) so they do not overwrite or break each other's files. Because they live in different neighborhoods, Windows must keep a list of all their addresses.

4. How Windows Searches the Path ("First Match Wins")

When you type a command, Windows reads your Path list from **left to right** (or top to bottom) and blindly opens the folders one by one to see if your required `.exe` file is sitting inside.

- **It is Instant:** Windows does not scan every single file slowly. It checks the folder's quick internal file index. It can search dozens of folders in a few milliseconds.

- **First Match Wins:** The absolute moment Windows opens a folder containing your target .exe file, it runs it and stops searching. It completely ignores any remaining folders down the line.
- **The Version Trap:** If you have two different versions of Python installed in two different folders, and both are on your Path, Windows will always run the one whose folder address appears **first** on your Path list.

5. Why Java Worked in IntelliJ but Not in MS-DOS

- **IntelliJ's Secret:** When you set up IntelliJ, it secretly downloads Java into its own private, isolated folder so it works perfectly inside the IDE app.
- **MS-DOS's Blindness:** IntelliJ never walked up to the Windows global Path cheat sheet to write down Java's new folder address.
- **The Fix:** To make Java work in MS-DOS, you simply have to copy that hidden Java folder address and append it to your Windows global Path list.